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Enhancing digital reading performance with a collaborative reading annotation system

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A B S T R A C T

As children now spend considerable time reading electronic media, digital reading skills and good reading comprehension are essential. However, many studies agree that screen-based reading leads to shallow reading, short attention spans, and poor comprehension. Therefore, this work presents a collaborative reading annotation system with a reading annotation and interactive discussion scaffold (CRAS-RAIDS) for improving reading performance in collaborative digital reading environments. This study used a quasi-experimental design. Fifty-three Grade 5 students were recruited from two classes of an elementary school in Taoyuan County, Taiwan. One class was randomly designated the experimental group used the proposed CRAS-RAIDS support for collaborative reading. The other class was designated the control group and used the traditional paper-based reading annotation method and face-to-face discussions. The two groups were then compared in terms of reading attitude, reading comprehension, and use of reading strategy in an active reading context. Analytical results show that the experimental group significantly outperformed the control group in direct and explicit comprehension, inferential comprehension performance, and use of reading strategy. Moreover, the experimental group, but not the control group, had a significantly improved reading attitude in the total dimensions and in the behavioral and affective sub-dimensions. Additionally, the experimental group showed positive interest and high learning satisfaction.

1. Introduction

Amazon, the online merchandising giant, now sells more e-books than paper books (Bounie, Eang, Sirbu, & Waelbroeck, 2012), and electronic resources at libraries are becoming increasingly popular (Liu, 2012). A recent survey by Gartner Inc. (2013) shows that the time people spend reading on a screen is now almost equal to the time spent reading printed text. The digital environment has substantially changed reading practices and behaviors (Liu, 2012). Eden and Eshet-Alkalai (2013) indicated that young readers of today are as proficient in reading digital text as they are in reading printed text because digital reading has become an everyday practice. The many notable advantages of digital reading over printed text include interactivity, nonlinearity, immediacy in accessing information, and the convergence of text, images, audio, and video. These features are absent in the print environment. However, Carr (2010) argued that screen reading and the fragmentary nature of hypertext reduce sustained reading and result in shallow reading. Liu (2005) also indicated that most of the time spent reading text on a screen reading is used for browsing and scanning, keyword spotting, one-time reading, non-linear reading, and reading selectively. Compared to paper-based reading, less time is spent on in-depth reading and concentrated reading. Wolf and Barzillai (2009) also encouraged further studies of the explicit instructions needed to achieve a deep comprehension processes and studying the formation of deep-reading processes for online reading. Early studies found that screen-based readers had lower comprehension compared to paper-based readers. However, this gap has narrowed in recent years (Cull, 2011). Specifically, in Eden and Eshet-Alkalai (2013), a study 93

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university students under active reading conditions confirmed that reading performance did not significantly differ between print and digital text. Their study supported the notion that digital reading has become an everyday practice and that users have gained digital proficiency. Since digital content is clearly growing, studies of the impact of reading on screen are extremely important.

The process of reading on a screen (non-linear reading, texts with hyperlinks, and shallow reading) tends to be cognitively different from the process of reading on paper (brain activation, the contextual environment, cognitive focus, comprehension, and reading speed) (Coiro & Dobler, 2007; Cull, 2011). Digital texts that incorporate hyperlinks and hypermedia introduce complications when defining comprehension because they require skills and abilities beyond those required for comprehension of conventional linear print (RAND Reading Study Group, 2002). However, many studies have asserted that the youth of today lack the ability to read deeply and to sustain engagement when reading online (Birkerts, 1994; Carr, 2010; Liu, 2005; Wolf & Barzillai, 2009). Fuchs et al. (2001) indicated that, compared to students with good reading skills, those with poor reading skills have lower self-esteem, more discipline problems, and a lower rate of school completion. Therefore, tools or strategies for assisting readers and for improving their online reading skills and performance are urgently needed.

Collaborative annotations of digital texts can accumulate rapidly and allow readers to share their knowledge. Annotations typically facilitate text review. Most importantly, annotated content can help readers obtain a deeper and broader understanding compared to digital content without annotations (Porter-O'Donnell, 2004). Many studies (Nokelainen, Miettinen, Kurhila, Floréen, & Tirri, 2005; Ovsiannikov, Arbib, & McNeill, 1999; Rau, Chen, & Chin, 2004) have demonstrated that collaborative annotation tools can promote reading performance and benefit collaborative reading. However, current collaborative annotation systems developed to improve reading comprehension do not provide an effective reading annotation and interactive discussion scaffold that can direct readers in annotating digital texts. Bull et al. (1999) indicated that a scaffold is an interactive process in which a teacher or facilitator assists a learner in building a "structure" to contain and frame new information. Additionally, a scaffold can be provided by teachers, peers, or computers and may include tutoring, performance systems, and reciprocal teaching. Providing a scaffold has been an effective strategy for helping students perform high-order cognitive activities, particularly reading (Chen, Teng, Lee, & Kinshuk, 2011; Clark & Graves, 2005). Additionally, various forms of collaborative reading are often included in discussion-based teaching methods (Kiili, Laurinen, Marttunen, & Leu, 2012). Notably, educators are increasingly using online asynchronous discussion tools for learning and instruction because these tools have fewer time and space restrictions than traditional face-to-face discussion (Chen & Chiu, 2008). Online asynchronous discussions also provide learners with opportunities to prepare, reflect, think, and search for additional information before participating in a discussion (Chen & Chiu, 2008). Wolf (2008) indicated that recent research on annotation interfaces provides convincing evidence that anchored, annotation-based discussion environments may deepen discussions of a text. Thus, an online asynchronous discussion that includes debate about digital texts with annotations can further help readers understand texts.

To enhance collaborative annotation applications in digital reading environments, this work presents a collaborative reading annotation system with a reading annotation and interactive discussion scaffold (CRAS-RAIDS) that improves the reading performance of learners by helping them apply annotation strategies and engage in interactive discussion in collaborative digital reading environments. This work focused on guiding students in annotating digital texts and in discussing the annotations since open-ended discussions are often ineffective and confusing to students (Michaels & Bruce, 1991). That is, the research purposes of this work are to determine whether collaborative reading with the proposed CRAS-RAIDS support and with traditional paper-based reading annotation and face-to-face discussion differ in terms of reading attitude, reading comprehension, and use of reading strategy in active reading context. Another objective is to determine whether collaborative reading using the proposed CRAS-RAIDS support increases web-based learning willingness and learning satisfaction.

2. Literature review

2.1. Digital reading support by the web-based reading annotation system

Reading is an activity characterized by different purposes and requires different skills in handling reading materials. That is, reading is a complex behavior (Liu, 2005). The recent advent of digital media and the growing number of available digital documents have profoundly changed reading practices (Liu, 2005). Specifically, new information and computer technology (ICT) offer alternative ways to support reading activities, and many reading assistance tools anticipate widespread change in reading behaviors. Traditionally, printed books are annotated with a pen or pencil; however, printed annotations are not as useful as knowledge stored in computers in terms of dissemination and sharing of knowledge via the Internet. Using a digital annotation tool to annotate digital texts can overcome these shortcomings. Many studies have confirmed the effectiveness of recently developed computer-assisted collaborative reading annotation systems for promoting reading performance (Chen, Chen, Hong, Liao, & Huang, 2012; Mendenhall & Johnson, 2010; Su, Yang, Hwang, & Zhang, 2010).

Su et al. (2010) proposed a personalized annotation management system (PAMS) that manages, shares, and reuses individual and collaborative annotations. The PAMS also includes a shared mechanism for discussions of annotations by multiple users. Analytical results show that the PAMS increases learning achievement in collaborative learning environments and that the effectiveness of annotations for improving learning achievement increases when the sharing mechanism is used. Steimle, Brdiczka, and Mühlhäuser (2009) developed the CoScribe system which can use a digital pen to annotate printed lecture slides for collaborative reading. They demonstrated that combining printed lecture slides with a digital pen can effectively support the annotation process and collaborative reading. Chen, Wang, and Chen (2014) combined a self-regulated learning (SRL) mechanism with a digital reading annotation system (DRAS) that Grade 7 students can use collaboratively to generate rich and high-quality annotations that promote their English-language reading performance. Compared with the learners who used the proposed DRAS without SLR support, the reading comprehension and annotation abilities of the learners who used the proposed DRAS supported by the SLR mechanisms significantly improved. Mendenhall and Johnson (2010) developed HyLighter, an online system that facilitates annotation sharing to foster development of critical thinking skills and reading comprehension in university undergraduates. Their study also demonstrated that HyLighter enhances reading comprehension, critical thinking and meta-cognition skills. Chen et al. (2012) developed a web-based collaborative reading annotation system that enhances knowledge sharing and promotes discussion among learners to improve their reading performance in a digital library environment. The annotated digital material provided useful knowledge to readers. Additionally, digital library content grows dynamically as readers contribute knowledge.

However, to the best of our knowledge, no study has focused on developing a collaborative reading annotation system with a reading annotation and interactive discussion scaffold for guiding readers in annotating digital texts and helping them to discuss these annotated texts.

2.2. Promoting reading performance through collaborative reading, reading scaffold, and computer-mediated discussions

Kiili et al. (2012) claimed that, like collaborative learning, collaborative reading is socially contextualized and requires at least one other person. Collaborative reading also includes a process in which meaning and knowledge are jointly constructed through text-based discussion. Their study explored collaborative reading as an activity with the potential for co-construction of meaning and knowledge. Huang (2012) also indicated that in collaborative reading, students become aware that reading is a group activity. By working and communicating with each other, readers develop new ideas, improve their reading skills, and reduce their tension and anxiety. Unlike individual reading, collaborative reading capitalizes on the sharing of resources and reading outcomes. The collaborative learning methods most commonly used in conventional reading instruction are Cooperative Integrated Reading and Composition and Reciprocal Teaching (Fuchs et al., 2001). However, traditional collaborative reading instruction is bounded by time and space and has limited opportunities for interaction, communication, coordination, negotiation, sharing, and interactivity within a learning group. Many recently developed computer-supported collaborative learning (CSCL) systems can explore the social nature of learning and focus exclusively on enhancing mediated collaboration among multiple learners and on facilitating reading instruction, particularly when developing collaborative reading annotation systems that promote reading performance in digital reading environments (Chen et al., 2012; Su et al., 2010; Yang, Zhang, Su, & Tsai, 2011). Reading annotations generally help learners by improving four skills: attention, organization, indexing, and discussion (Yang et al., 2011). Collaborative strategic reading (CSR) is also widely used for collaborative reading. Notably, in CSR, students can apply specific strategies, including brainstorming and prediction (*preview*), monitor understanding (*click and clunk*), find the main idea (*get the gist*), and generate questions and review key ideas (*wrap up*), to enhance their reading comprehension; the strategies are often used in small cooperative groups in which each student plays a critical role and this role is associated with effective functioning of the group and the implementation of strategies (Klingner, Vaughn, Arguelles, Hughes, & Leftwich, 2004).

Yelland and Masters (2007) identified three scaffolds, which in their work are called cognitive, technical and affective; these can be conceptualized to enhance learning performance. The cognitive scaffold denotes activities related to the development of conceptual and procedural understandings which involve either techniques or devices that assist learners. The technical scaffold is related to the use of computers. The affective scaffold comprises the mechanisms that help learners focus on a task and encourage them to apply higher levels of thinking and operating when engaged in various learning activities. In one study of associated reading practices with reading scaffold support, Chen et al. (2011) combined QR (Quick Response) codes with mobile technology to deliver supplementary materials and scaffolded questions to support paper-based reading. Their experimental results suggested that using QR codes for directly accessing supplementary materials did not significantly improve reading comprehension; however, the use of a scaffolded questioning strategy significantly improved the understanding of a text. Li, Chen, and Yang (2013) developed the visual cue map, which presents pages and within-page spatial cues in an interactive toolbar and reflects the physical structure of the e-book and the relative relationship between cues and pages; the goal was to improve e-book reading and navigation. Participants who used the e-book system with the visual cue map completed ten navigational tasks significantly faster and scored significantly higher compared to participants who used the e-book system without the visual cue map. Moreover, Jin (2013) developed visual structure design guidelines for using computer screenshots to enhance comprehension of text structures and developed selective attention design guidelines for presenting key phrases to maintain the attention of learners to essential content. These guidelines were based on psychological, instructional, technological foundations that can affect the visual design of digital text. Their experimental results showed that the visual structure and selective-attention design guidelines improved the content comprehension, the structural comprehension, and the content usability of digital texts. Visual cue maps or visual structure design guidelines can be considered as scaffolds for supporting the application of reading skills.

Moreover, discussion-based teaching methods are often applied in various collaborative reading strategies to develop skills in interpretation, high-order questioning, and explorations and argumentation, which are particularly beneficial in promoting reading performance (Kiili et al., 2012). Generally, a discussion can be a face-to-face conversation (Chiu, 2008) or a computer-mediated discussion (discussion board, online forums, and chat rooms) (Chen & Chiu, 2008). Despite the importance often assigned to participation in classroom discussions, many studies have found that students rarely or never ask questions or respond to questions in the classroom (Caspi, Chajut, Saporta, & Beyth-Marom, 2006; Crombie, Pyke, Silverthorn, Jones, & Piccinin, 2003). Guiller, Durndell, and Ross (2008) reported that, compared with face-to-face classroom discussion, computer-mediated discussion increases the use of formal, research-based evidence and critical thinking quality because it is asynchronous, allowing considerable time for reflection before responding. Additionally, the process of expressing ideas in words may benefit rhetorical and writing skills. Computer-mediated discussion also has fewer time and space restrictions and allows more opportunities for learners to prepare, reflect, think, and search for extra information before participating in a discussion (Chen & Chiu, 2008). Most importantly, disagreements occur more often during computer-mediated discussions than in face-to-face discussions because computer-mediated discussions are often de-personalized and participants are generally not concerned about the feelings of others (Chen, 2004); according to Chiu and Khoo (2003), new ideas presented in online discussions are often met with disagreement. In contrast, since messages that agree with a previous message do not require additional content, they provide little added information for others to reference. Based on the positive effects of collaborative reading, reading scaffold, and interactive discussion in reading practices, this work applied CRAS-RAIDS to support digital reading and assessed its potential in promoting reading attitude, use of reading strategy use, and reading comprehension.

2.3. Effects of reading attitude and reading strategy use on reading comprehension

With advances in ICT, paper books are giving way to digital media. This growth is significantly changing reading practices. The RAND Reading Study Group (2002) defined reading as “the process of simultaneously extracting and constructing meaning through interaction and involvement with written language.” According to the National Institute of Child Health and Human Development (2000), comprehension occurs “when readers actively relate the ideas represented in print to their own knowledge and experiences and construct mental representations in memory." Reading for understanding is a challenging task, particularly when a reader is unfamiliar with the material. Compared to a printed text, many studies have demonstrated that reading a digital text leads to less comprehension than when reading paper-based text (Carr, 2010; Liu, 2005; Morineau, Blanche, Tobin, & Guéguen, 2005). Carr (2010) found that comprehension of electronic texts containing hyperlinks decreased as the number of links increased. However, a study of 66 US college students by Moyer (2011) found no significant differences among printed text, e-books, and audio books. Eden and Eshet-Alkalai (2013) similarly found that the reading performance of 93 university students did not significantly differ between print and digital reading under active reading conditions. Clearly, the comprehension gap between reading on a screen *versus* reading on paper is gradually decreasing because digital reading has become an everyday practice.

Moreover, many studies indicated that reading attitude and reading strategy are two relatively important factors that are strongly related to reading comprehension (Brown, Campione, & Day, 1981; Coiro & Dobler, 2007; Kidwai, 2009; Petscher, 2010). Reading attitudes have long been considered an important psychological construct because of their important role in moderating motivation to read and intention to read and in mediating the relationship between individual beliefs and reading activities (Petscher, 2010). Alexander and Cobb (1992) emphasized that good reading attitudes are a prerequisite for reading and that reading attitude and reading comprehension are strongly related. Most teachers agree that attitude significantly affects reading achievement (Russ, 1989). In Petscher (2010), a meta-analysis of 32 studies of attitudes and achievement found that reading attitudes and achievement are moderately related ($Zr = 0.32$). Since positive attitudes toward reading tend to generate high reading achievements, this study investigated whether digital reading using the proposed CRAS-RAIDS enhances reading attitude.

Brown et al. (1981) identified two general difficulties that preclude reading effectiveness: inadequate background knowledge and poor reading strategies. Reading comprehension can be improved by applying reading strategies in real time. Strategies applied in printed text by expert readers include previewing, setting goals, making predictions, monitoring understanding, asking questions, and interpreting (Coiro & Dobler, 2007). Kidwai (2009) proposed five reading strategies for a Web-based learning environment: text-macrostructure or chunking strategy, summarization strategy, imagery strategy, reading self-assessment or comfort-meter strategy, and note-taking strategy. The summarization strategy requires readers to write a summary for each section whereas the note-taking strategy requires readers to take notes on an element-by-element basis. The notes are then attached to the elements that make up an instructional unit. However, skilled readers often integrate several processes to aid comprehension. Pugh (1978) proposed that scanning, search reading, skimming, receptive reading, and responsive reading are the five most common reading strategies applied by students, and each technique requires a different level of intellectual engagement with the content. Responsive reading, in which the reader annotates the text, is also common. Moreover, proficient readers can skillfully apply sophisticated reading strategies and can understand how a text is organized. They also have more metacognitive skills compared to less proficient readers (Yau, 2005). Since the use of reading strategies affects reading comprehension, this study examines whether collaborative reading with the proposed CRAS-RAIDS support promoting use of reading strategy.

3. Research methodology

3.1. Research variables

This work examines differences in reading attitude, reading comprehension, and use of reading strategy in an active reading context between collaborative reading with the proposed CRAS-RAIDS support and collaborative reading with traditional paper-based reading annotations and face-to-face discussion. Therefore, the independent variable was either collaborative reading with CRAS-RAIDS support or collaborative reading with conventional paper-based reading annotation and face-to-face discussion for two selected Progress in International Reading Literacy Study (PIRLS) articles. Dependent variables are reading attitude, reading comprehension, and the use of reading strategies. Reading attitude is assessed using three components—*cognition*, *affection*, *and behavior*; pretest and posttest scores of the PIRLS reading comprehension test indicate reading comprehension performance; use of reading strategy is assessed based on Retrieving and Straightforward Inferencing and Interpreting, Integrating, and Evaluating in PIRLS; and the control variables are using the same PIRLS reading articles and the same experimental period for the two different treatments.

3.2. Experimental design

This quasi-experimental design used in this study included a nonequivalent control group was used because of the difficulty of randomly selecting participants in actual teaching scenarios. Study participants were recruited randomly from Grade 5 students in two classes at an elementary school in Taoyuan County, Taiwan. Participants were randomly assigned to either the experimental group or the control group. Students in the experimental group performed collaborative reading with the proposed CRAS-RAIDS support for two selected PIRLS articles while the control students performed collaborative reading with traditional paper-based reading annotations and face-to-face discussion for the same PIRLS articles. Learners in the experimental group were allowed to make their annotations private or public; private annotations were displayed only on the personal webpage whereas public annotations were shared with their peers. However, students were encouraged to make public rather than private annotations. Moreover, learners in the experimental group can select any digital text to annotate. These annotations can be determined an appropriate type of reading annotation tag based on seven different reading annotation scaffolds provided in the collaborative reading annotation system. Additionally, learners in the experimental group were allowed to browse annotations by their peers and to respond to these annotations based on six interactive discussion scaffolds provided in the collaborative reading annotation system. Conversely, learners in the control group were asked to share their annotations in writing and in face-to-face discussions. Any number of students was allowed to participate in face-to-face discussions, and control group learners were asked to record their discussions, including names of participators and discussion content.

3.3. Experimental procedures

Before the collaborative reading activities, a reading comprehension pretest sheet of the first selected PIRLS article and a reading attitude scale were used to compare prior knowledge and the initial reading attitude between the two groups. Before starting the reading program, both groups received a 1-h training session in how to annotate conventional paper-based texts, conduct face-to-face discussions, and operate the proposed CRAS-RAIDS for reading activities. Both groups then began collaboratively reading the first selected PIRLS article. The collaborative reading activities included writing autonomous annotations based on the points of view of individuals and participating in interactive discussions of annotations. After performing three collaborative reading activities within a week for the first PIRLS article, both groups took the reading comprehension posttest. To ensure the reading comprehension pretest and posttest sheets of the first PIRLS article have the same difficulty, the pretest and posttest sheets used in the study were the same. However, to consider that performing the reading comprehension pretest of the first PIRLS article for assessing prior knowledge of the two groups may lead to learners in both groups to search their textbooks or the Internet for the pretest sheet answers of the first PIRLS article during learning processes, which adversely affecting the accuracy of the reading comprehension posttest, learners in both groups were asked to perform the collaborative reading activity for the second selected PIRLS article. To avoid exposing the pretest sheet, the pretest sheet for the second article was not thus administered at this stage. Similarly, after performing three 30-min collaborative reading activities within a week for the second PIRLS article, both groups took the reading comprehension posttest for the second PIRLS article and the posttest for reading attitude. No students were directed by a teacher during the collaborative reading activity. That is, all learners performed autonomous learning in an active learning context. After completion of all collaborative reading activities, reading attitude, reading comprehension, and use of reading strategy were compared between the two groups.

3.4. Research participants

The participants were 53 Grade 5 students aged 10–11. One class of 28 students (13 males and 15 females) was randomly selected as the experimental group while the remaining class of 25 students (14 males and 11 females) was designated the control group.

3.5. Research instruments

3.5.1. The developed CRAS-RAIDS

This developed system has a user-friendly interface that readers can use for annotating digital texts and discussing digital annotations. The main functionalities of the system are as follows.

3.5.1.1. Reading annotation scaffold for annotated digital texts. The proposed system allows readers to create, modify, and delete annotations from digital texts. Fig. 1 shows the user interface for annotating texts. Users can make private or public annotations. Public annotations are shared with other readers whereas private annotations are only displayed on the personal annotation webpage for each student. Moreover, to help users create annotations with appropriate semantic tags to promote reading comprehension, the proposed system provides seven types of reading annotation scaffolds: reasoning, discrimination, linking, summary, quizzing, explanation, and other (Fig. 1). These scaffolds help users to identify appropriate annotation types. The functions of the annotation scaffolds are as follows.

(1) Reasoning: Integrating and interpreting the emphasis of an annotated digital text.
(2) Discrimination: Asserting distinct viewpoints based on how students evaluate and examine the meaning of an annotated text.

Fig. 1. The user interface of reading annotation scaffold for annotating texts in the collaborative reading annotation system.

(3) Linking: Connecting knowledge, life experiences, or articles with annotated digital texts.
(4) Summary: Summarizing the meaning of an annotated digital text based on key information or concepts extracted from the annotated digital text.
(5) Quizzing: Expressing doubts and thoughts about an annotated digital text.
(6) Explanation: Obtaining supplementary explanations or instances for an annotated text.
(7) Other: Determining the type of annotated digital text when the annotated digital text cannot be classified as a predefined annotation scaffold.

3.5.1.2. Interactive discussion scaffold for annotated digital texts. Fig. 2 shows the user interface of the interactive discussion scaffold used to respond to annotated texts in the collaborative reading annotation system. The scaffold provides readers with a space to discuss annotations with other readers. Interactive discussions enable readers to respond to, clarify, and debate annotations contributed by other readers. To help readers create structured and meaningful interactive discussions that promote comprehension, the proposed system provides six interactive discussion scaffolds: reasoning, discrimination, quizzing, clarification, debugging, and other. These scaffolds help users identify the discussion type. The functions of the scaffolds are as follows.

(1) Reasoning: Making reasoning annotations in response to a digital text.
(2) Discrimination: Making discrimination annotations in response to a digital text.
(3) Quizzing: Making quizzing annotations in response to a digital text.
(4) Clarifying: Clarifying opinions or thoughts about annotated digital texts.
(5) Debugging: Revising incorrect concepts, opinions or thoughts about an annotated digital text.
(6) Other: Using a classification other than the pre-defined classifications for a discussion.

3.5.1.3. Favorite annotation. The proposed system includes a function that allows readers to mark their favorite annotations by clicking the love icon. Favorite annotations are very helpful when reviewing annotations of individual readers.

3.5.2. Conventional paper-based reading annotations

Currently students typically use a pencil or pen to annotate print texts. Hand-written annotations may include highlighting, underlining, or making comments, footnotes, tags, and links in the margins of pages. Readers frequently highlight or underline words, phrases, or passages, write short comments within margins or between lines, or use long notes in blank spaces or near figures to add complementary information. Fig. 3 shows conventional hand-written paper-based reading annotations made by subjects in the control group.

3.5.3. Reading comprehension assessment

The PIRLS, which is designed to measure children's reading literacy achievement, to provide a baseline for future studies of trends in achievement, and to gather information about children's home and school experiences in learning to read, is an international study of reading achievement in fourth graders (Martin, Mullis, & Kennedy, 2007). The PIRLS test sheets and scoring guidelines were used to assess

Fig. 2. The user interface of the interactive discussion scaffold for responding the texts with annotation contents in the collaborative reading annotation system.

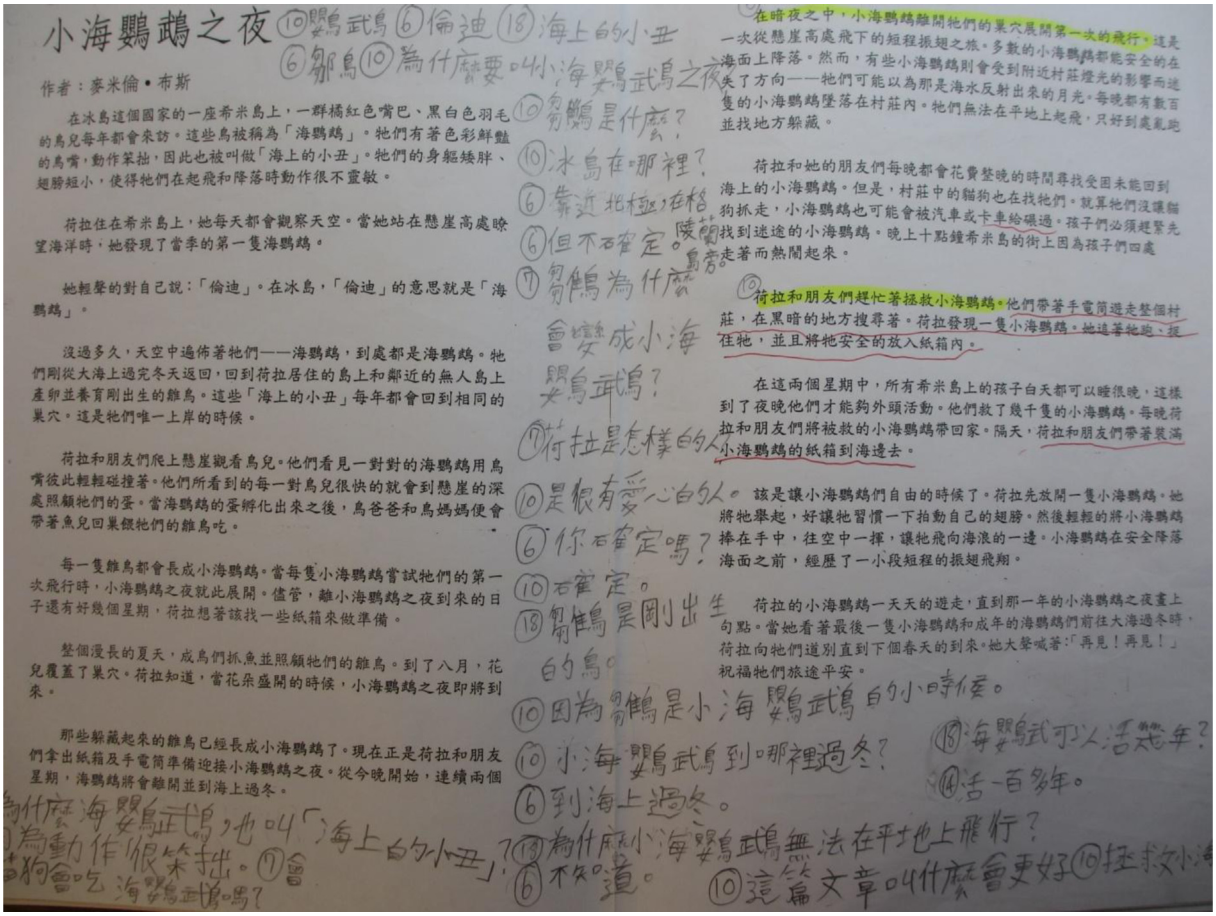
小海鸚鵡之夜
作者：麥米倫・布斯

Fig. 3. Examples of hand-written paper-based reading annotations made by the control group students.

comprehension of the two selected PIRLS articles in the study. The two PIRLS articles were "An unbelievable night" and "Puffings." To assess reading comprehension, PIRLS measures reading comprehension in two parts: retrieving and straightforward inference; and interpreting, integrating, and evaluating. Retrieving and straightforward inference were termed direct and explicit comprehension; interpreting, integrating, and evaluating were termed inferential comprehension. Each part comprises about half of the assessment items. The direct and explicit comprehension assessment includes items for assessing the retrieval process and those for assessing straightforward inference. Conversely, the inferential assessment combines the interpreting and integrating items with examining and evaluating items. For example, the test sheet for the PIRLS article "An unbelievable night" is consisted of 6 multiple-choice items, 1 fill-in-the-blank item, and 5 short-answer items. The correct answer for the multiple-choice item can get 1 point, but the rating scores for the fill-in-the-blank and short-answer items have to refer the scoring guidelines provided for the PIRLS article. The scoring guidelines simultaneously provide the detailed explanations to direct how to rate score according to the answer of examinee and example answer for some short-answer item. Basically, the scoring guidelines direct raters to determine the score based on the completeness or level of an answer, such as 3 points for extensive comprehension, 2 points for complete or satisfactory comprehension, 1 point for partial comprehension or acceptable response. The reading comprehension performance is determined by summing all the scores in the test sheet for the PIRLS article.

3.5.4. Reading strategy assessment

To assess the use of reading strategy, ten reading experts, each of whom had over 10 years of experience in reading instruction in Taiwan's primary school, were invited to determine the assessment scores of eight frequently used reading strategies based on levels of reading skill. The ten reading experts were asked to rate a score ranging from 1 for "the reading strategy with the lowest reading skill" to 4 for "the reading strategy with the highest reading skill" in the eight considered reading strategies. Table 1 shows the results. Based on the average rating score of ten reading experts, the assessment score of each considered reading strategy was determined as an integer score. Similarly, two elementary school teachers, each of whom had over 10 years of experience in reading instruction, were invited to identify the category of the used reading strategy for each reading annotation and discussion post. Intercoder reliability based on Cohen's kappa was as high as 0.92. The use of reading strategy of each learner is determined by summing all the scores of the used reading strategies in the reading annotations and discussion posts contributed by the learner during reading activities.

3.5.5. Reading attitude assessment

Reading attitude was measured using a 13-item scale that assessed three dimensions—*cognition*, *affection*, *and behavior* (Anderson, 1988). The cognitive dimension measures beliefs or thoughts about situations and events; the affective dimension measures feelings about a situation or events; the behavioral dimension measures behavioral intention under a particular context. Responses were made on a

Table 1
The assessment scores of eight frequently used reading strategies determined by ten reading experts.

Item	The considered reading strategy							
	Explanation	Debugging	Quizzing	Clarifying	Linking	Summary	Reasoning	Discrimination
Expert 1	1	2	1	2	3	3	4	4
Expert 2	1	2	2	2	3	3	4	4
Expert 3	1	2	1	2	2	3	4	4
Expert 4	1	2	1	2	3	3	4	4
Expert 5	1	2	1	2	3	3	4	4
Expert 6	1	2	1	1	2	3	4	4
Expert 7	1	2	1	2	3	3	4	4
Expert 8	1	1	2	2	3	3	4	4
Expert 9	1	1	1	2	2	3	4	4
Expert 10	1	2	2	2	2	3	4	4
Average rating score	1	1.8	1.3	1.9	2.6	3	4	4
The determined score of reading strategy use	1	2	1	2	3	3	4	4

4-point Likert-type scale ranging from 1 for “disagree” to 4 for “strongly agree.” Scale reliability was examined by the 97 other students. Cronbach alpha values for the *cognition*, *affection*, *and behavior* subscales were 0.805, 0.692, and 0.820, respectively, and the Cronbach alpha value for the total scale was 0.845, implying that the scale has high reliability and sensitivity for reading attitude.

3.5.6. Learning satisfaction assessment

A learner satisfaction questionnaire was also designed to assess four dimensions of learning satisfaction in the experimental learners. The four dimensions were learning activity, interactive behavior, learning environment, and learning system. The students responded to each questionnaire item using a 6-point Likert-type scale ranging from 1 for “strongly disagree” to 6 for “strongly agree.” The Cronbach's alpha values of the four dimensions are 0.859, 0.894, 0.932, and 0.868, and the Cronbach's alpha of the total questionnaire is 0.954, implying that the questionnaire has a high reliability and sensitivity to learning satisfaction.

4. Experimental results

4.1. Analysis of discussion posts for both groups

To determine whether discussion behaviors of both groups significantly differed, two elementary school teachers, each of whom had over 10 years of experience in reading instruction, were invited to categorize each discussion post based on the defined interactive discussion scaffold. Intercoder reliability based on Cohen's kappa was as high as 0.95. Table 2 shows the distribution of types of discussion posts and the independent-sample *t*-test results of the number of discussion posts for the two groups. This work found that the total number of discussion posts in the experimental group for the first PIRLS article is 321 and the second PIRLS article is 582. The total number of discussion posts in the control group for the first PIRLS article is 140 and the second PIRLS article is 166. The total number of discussion posts in the experimental group was almost two times higher for the second article compared to the first article; however, the number of discussion posts in the control group was similar. The results show that the willingness of readers who participate in discussions increased in the experimental group for the second PIRLS article but not in the control group. Interestingly, this work found that discussion posts by the control group focused mainly on quizzing, clarification, and other types as well as most discussion posts of other type belong to chats, which are irrelevant with the subject of the reading article. That is, the processes of face-to-face discussion were bounded in the cycle of proposing questions and clarifying, and high percentage of discussion posts tended to diverge from the subject of the reading article. Conversely, in addition to the quizzing and clarification types, most discussion posts by the experimental group were discrimination, reasoning, and other types. However, compared to the control group, lower percentage of discussion posts in the experimental group belongs to chats, which are irrelevant with the subject of the reading article. A content analysis by Stemler (2001) indicated that discrimination and reasoning when reading are high-level cognitive abilities. Therefore, the proposed CRAS-RAIDS not only facilitates high-level thinking and interactive discussions, it also helps learners focus on discussing the subject of the reading article.

Moreover, this work performed an independent-sample *t*-test to assess whether the number of discussion posts significantly differed between the two groups. Analytical results revealed significantly more posts by the experimental group in both articles ($t = 2.077$, $p < .05$; $t = 4.124$, $p < .05$) (Table 2). Clearly, the experimental group, which performed collaborative reading with the proposed CRAS-RAIDS, was more willing to participate and was more active in interactive discussions compared to the control group, in which collaborative reading was performed using conventional paper-based reading annotations and face-to-face discussion.

4.2. Reading comprehension for both groups

To assess the reading comprehension performance of both groups, the paired-sample and Analysis of Covariance (ANCOVA) were applied. Prior knowledge was measured by the pretest scores for the first PIRLS article, and reading comprehension was measured by the posttest scores for the first and second PIRLS articles. Reading comprehension was assessed and compared in terms of direct explicit comprehension and inferential comprehension.

Table 2
Distribution of types of discussion posts and independent-sample t-test results of the number of discussion posts for the two groups.

	The first selected PIRLS article		The second selected PIRLS article	
	The control group	The experimental group	The control group	The experimental group
The type of discussion posts				
Reasoning	11.8%	29.3%	3.5%	15.2%
Discrimination	0.7%	18.1%	0.7%	11.1%
Debugging	0%	0.5%	0.7%	0.9%
Clarification	26.5%	11.2%	32.2%	33.3%
Quizzing	31.5%	30.8%	34.1%	30.2%
Other	29.4%	10.2%	28.8%	9.3%
Comparison item				
Number of students	25	28	25	28
Total number of discussion posts	140	321	166	582
Mean	5.60	11.46	6.64	20.79
Std.	4.983	13.975	3.695	17.725
t	2.077*		4.124***	
Sig. (two tailed)	0.045		0.000	

* indicates $p < .05$; *** indicates $p < .001$.

Table 3
Descriptive statistics results of direct and explicit comprehension for the two groups.

Item	Leaning group	Number of students	Mean	Std.
The pretest of the first selected PIRLS article	The control group	25	76.12	17.812
	The experimental group	28	77.18	14.420
The posttest of the first selected PIRLS article	The control group	25	82.80	15.028
	The experimental group	28	83.50	12.524
The posttest of the second selected PIRLS article	The control group	25	58.76	16.754
	The experimental group	28	70.75	16.750

4.2.1. Analysis of direct and explicit comprehension for both groups

Table 3 shows the descriptive statistics results of direct and explicit comprehension for the two groups. First, the paired-sample t-test was used to determine whether direct and explicit comprehension by both groups differed significantly based on pretest and posttest scores for the first article. Analytical results revealed that direct and explicit comprehension significantly differed in the experimental group ($t = -2.250$, $p < .05$), but not in the control group ($t = -1.886$, $p > .05$) (Table 4). That is, direct and explicit comprehension was promoted in the experimental group of students who used CRAS-RAIDS, but not in the control group. Additionally, this work examined whether the posttest of direct and explicit comprehension of the first article in the experimental group was superior to that of the control group by ANCOVA. The first step is to analysis the homogeneity of regression coefficients. The F test result ($F = 0.005$, sig of $F = 0.943$) does not reach the significant level, thus it means the regression slope of two groups is equivalent. This result confirms the assumption of homogeneity of regression coefficients, and so this study further preceded the ANCOVA. The ANCOVA result does not reach the significant level ($F = 0.009$, sig of $F = 0.926$) after adjusting the dependent effect (group) with respect to the covariance (pretest). This result indicates that there is no statistically significant difference between the posttest of two groups, both learning modes obtained equivalent performance in terms of direct and explicit comprehension for the first article. However, posttest scores for the second article revealed that direct and explicit comprehension in the experimental group was superior to the control group ($F = 6.745$, sig of $F = 0.012$).

4.2.2. Inferential comprehension for both groups

Table 5 shows the descriptive statistics results of inferential comprehension for the two groups. Similarly, the paired-sample t-test was used to compare pretest and posttest scores for inferential comprehension for the first article (Table 6). The comparison revealed significant changes in both groups ($t = -4.175$, $p < .05$; $t = -5.157$, $p < .05$). In other words, the experimental and control group students performed well in terms of inferential comprehension. Moreover, the ANCOVA was also used to determine whether inferential comprehension between the two groups has significant difference. The first step is to analysis the homogeneity of regression coefficients. The F test result ($F = 1.417$,

Table 4
Paired-sample t-test results of direct and explicit comprehension for the two groups.

Leaning group	Item	Number of students	Mean	Std.	t	Sig. (two tailed)
The control group	The pretest of the first selected PIRLS article	25	76.12	17.812	−1.886	0.071
	The posttest of the first selected PIRLS article	25	82.80	15.028		
The experimental group	The pretest of the first selected PIRLS article	28	77.18	14.420	−2.250*	0.033
	The posttest of the first selected PIRLS article	28	83.50	12.524		

* indicates $p < .05$.

Table 5
Descriptive statistics results of inferential comprehension for the two groups.

Item	Leaning group	Number of students	Mean	Std.
The pretest of the first selected PIRLS article	The control group	25	61.72	22.094
	The experimental group	28	64.79	19.232
The posttest of the first selected PIRLS article	The control group	25	77.12	21.524
	The experimental group	28	86.68	16.097
The posttest of the second selected PIRLS article	The control group	25	70.36	27.983
	The experimental group	28	90.89	16.551

Table 6
Paired-sample t-test results of inferential comprehension for the two groups.

Leaning group	Item	Number of students	Mean	Std.	t	Sig. (two tailed)
The control group	The pretest of the first selected PIRLS article	25	61.72	22.094	-4.175^{***}	0.000
	The posttest of the first selected PIRLS article	25	77.12	21.524		
The experimental group	The pretest of the first selected PIRLS article	28	64.79	19.232	-5.157^{***}	0.000
	The posttest of the first selected PIRLS article	28	86.68	16.097		

*** indicates $p < .001$.

sig of $F = 0.240$) does not reach the significant level, thus it means the regression slope of two groups is equivalent. This result confirms the assumption of homogeneity of regression coefficients, and so this study further preceded the ANCOVA. The ANCOVA result does not reach the significant level ($F = 3.335$, sig of $F = 0.074$) after adjusting the dependent effect (group) with respect to the covariance (pretest). This result indicates that there is no statistically significant difference between the posttest of two groups, both learning modes obtained equivalent performance in terms of inferential comprehension for the first article. However, posttest scores for the second article revealed that inferential comprehension in the experimental group was superior to the control group ($F = 11.612$, sig of $F = 0.001$).

4.3. *Reading strategy use by both groups*

The independent-sample t-test was used to determine whether the groups significantly differed in the use of reading strategies assessed by two elementary school teachers who had over ten years of reading instruction experience (Table 7). The use of reading strategies was significantly higher in the experimental group for both the first and second articles ($t = 3.003$, $p < .05$; $t = 4.545$, $p < .05$).

4.4. *Reading attitude for both groups*

The reading attitude scale was used to compare reading attitude after the different learning treatments. Table 8 shows the descriptive statistics results of reading attitude for the two groups. First, the paired-sample t-test was used to detect significant changes in pretest and posttest scores for reading attitude in each group (Table 9). In the experimental group, reading attitude significantly improved in terms of the behavioral and affective dimensions and total scale ($t = -3.47$, $p < .05$; $t = -3.57$, $p < .05$; $t = -2.96$, $p < .05$). The control group showed no significant improvements in the three dimensions or in the total scale. Moreover, comparison of reading attitude after learning based on the ANCOVA showed that the three dimensions and total scale did not significantly differ in the two groups. Restated, final reading attitude did not significantly differ in the two groups.

4.5. *Learning satisfaction in the experimental group*

Learning satisfaction was then assessed in the experimental group (Table 10). The average scores for learning activity, interactive behavior, learning environment, and learning system were 5.21, 5.28, 5.23, and 5.43, respectively. Restated, most students in the experimental group "agreed" or "strongly agreed" that using the proposed CRAS-RAIDS positively affected the four dimensions. Specifically, the average response to the statement "The annotations contributed by other readers are very helpful to my learning" was 5.43 (standard deviation, 0.63); the average response to the statement "The CRAS-RAIDS facilitates interactive discussion with other readers" was 5.5 (standard deviation, 0.64); the average response to the statement "I will read the reading annotations shared by other readers on CRAS-RAIDS" was 5.64 (standard deviation, 0.55); the average response to the statement "The functionalities in the CRAS-RAIDS for responding discussion issues is very easy to use" was 5.71 (standard deviation, 0.53); the average response to the statement "Browsing annotations by

Table 7
Independent-sample t-test results of reading strategy use for the two groups.

The selected PIRLS article	Leaning group	Number of students	Mean	Std.	t	Sig. (two tailed)
The first article	The control group	25	9.20	7.714	3.003^{**}	0.005
	The experimental group	28	24.86	26.356		
The second article	The control group	25	10.52	6.539	4.545^{***}	0.000
	The experimental group	28	45.39	40.002		

** indicates $p < .01$; *** indicates $p < .001$.

Table 8
Descriptive statistics results of reading attitude for the two groups.

Leaning group	Item	Number of students	Mean	Std.
The control group	The pretest of cognitive dimension	25	3.21	0.636
	The posttest of cognitive dimension	25	3.31	0.543
	The pretest of behavioral dimension	25	3.1100	0.56862
	The posttest of behavioral dimension	25	3.1000	0.62082
	The pretest of affective dimension	25	3.5867	0.53817
	The posttest of affective dimension	25	3.76	0.297
	The pretest of total scale	25	3.2677	0.45946
	The posttest of total scale	25	3.3508	0.40950
The experimental group	The pretest of cognitive dimension	28	3.33	0.521
	The posttest of cognitive dimension	28	3.34	0.566
	The pretest of behavioral dimension	28	3.1518	0.50616
	The posttest of behavioral dimension	28	3.3482	0.51523
	The pretest of affective dimension	28	3.5595	0.40626
	The posttest of affective dimension	28	3.82	0.212
	The pretest of total scale	28	3.3297	0.38425
	The posttest of total scale	28	3.4533	0.39130

Table 9
Paired-sample t-test results for reading attitude in each group.

Leaning group	Item	Number of students	Mean	Std.	t	Sig. (two tailed)
The control group	The pretest of cognitive dimension	25	3.21	0.636	−1.095	0.284
	The posttest of cognitive dimension	25	3.31	0.543		
	The pretest of behavioral dimension	25	3.1100	0.56862	−1.095	0.284
	The posttest of behavioral dimension	25	3.1000	0.62082		
	The pretest of affective dimension	25	3.5867	0.53817	0.092	0.927
	The posttest of affective dimension	25	3.76	0.297		
	The pretest of total scale	25	3.2677	0.45946	−1.256	0.221
	The posttest of total scale	25	3.3508	0.40950		
The experimental group	The pretest of cognitive dimension	28	3.33	0.521	−0.08	0.937
	The posttest of cognitive dimension	28	3.34	0.566		
	The pretest of behavioral dimension	28	3.1518	0.50616	−3.47**	0.002
	The posttest of behavioral dimension	28	3.3482	0.51523		
	The pretest of affective dimension	28	3.5595	0.40626	−3.57**	0.001
	The posttest of affective dimension	28	3.82	0.212		
	The pretest of total scale	28	3.3297	0.38425	−2.96**	0.006
	The posttest of total scale	28	3.4533	0.39130		

** indicates $p < .01$.

my peers in the CRAS-RAIDS is very easy" was 5.61 (standard deviation, 0.79). These high scores for learning satisfaction and their low standard deviations show that most students in the experimental group were satisfied with the proposed CRAS-RAIDS and that their opinions were relatively consistent. More importantly, the average response to the statement "I will continue using the CRAS-RAIDS to perform reading activity in the future" was 5.5. These high scores confirm that the proposed CRAS-RAIDS is a well-designed assistive reading tool and significantly improves digital reading performance.

Moreover, Pearson product–moment correlation was applied to analyze the correlations among learning satisfaction, direct and explicit comprehension performance, and inferential comprehension performance for learners in the experimental group who collaboratively read two PIRLS articles with CRAS-RAIDS support. Analytical results show that no significant correlations existed among learning satisfaction, direct and explicit comprehension performance, and inferential comprehension performance for learners in the experimental group. The work inferred that the possible reason is that the reading comprehension performance of learners in the experimental group not only is affected by their individual efforts on reading activities, but also is promoted by collaborative reading annotations from peers, such that learning satisfaction and reading comprehension performance were uncorrelated.

5. Discussion

Early studies revealed low comprehension levels for digital texts (Carr, 2010; Liu, 2005); however, recent studies indicate that the reading comprehension gap between digital texts and paper texts has gradually decreased as digital reading becomes common and as digital readers gain reading proficiency (Cull, 2011; Eden & Eshet-Alkalai, 2013). Eden and Eshet-Alkalai (2013) compared print and digital texts in terms of comprehension under active reading conditions in 93 university students and found no significant difference. Moreover, many studies argued that the generation that has grown up in this digital era lacks the ability to read deeply and sustain prolonged attention when reading online (Birkerts, 1994; Carr, 2010; Liu, 2005; Wolf & Barzillai, 2009). The current study, however, confirms that using the proposed CRAS-RAIDS for online collaborative reading significantly improves both direct and inferential comprehension. These analytical results are encouraging because comprehension is the ability to read between the lines while making connections not explicitly stated in a text (Beck, 1989). This understanding is considered central to skilled reading (Garnham & Oakhill, 1996). Restated, the experimental group not only had superior direct comprehension in shallow reading, but also had inferential comprehension in deep reading. Moreover, Yelland and Masters

Table 10
Assessment of learner satisfaction in the experimental group. The significance of bold values indicates the corresponding item has high and consistent satisfactory degree.

	Mean	Std.
Learning activity		
01. The difficulty level of the selected reading articles on the CRAS-RAIDS is moderate	4.93	1.33
02. The annotations contributed by other readers are very helpful to my learning	**5.43**	**0.63**
03. The CRAS-RAIDS facilitates interactive discussion with other readers	**5.5**	**0.64**
04. Performing the designed reading activity on the CRAS-RAIDS is very challenging	4.93	1.01
05. The autonomous reading activity in CRAS-RAIDS is effective	5.11	0.83
06. Performing the designed reading activity in CRAS-RAIDS is very interesting	5.04	1.07
07. Using CRAS-RAIDS to perform the reading activity improves reading comprehension	5.21	0.83
08. Using CRAS-RAIDS to perform reading activities can improve skills in using reading strategies	5.36	0.73
09. The CRAS-RAIDS provides opportunities to practice reading strategies	5.32	0.72
10. Overall, the CRAS-RAIDS can help me improve my reading performance	5.29	0.76
Overall satisfaction with learning activity	5.21	
Interactive behavior		
11. I am willing to share my reading annotations with other readers through the CRAS-RAIDS	5.32	0.82
12. I will read the reading annotations shared by other readers on CRAS-RAIDS	**5.64**	**0.55**
13. I will try to participate in discussions on CRAS-RAIDS	5.18	0.86
14. I will respond when other readers propose discussion issues	5	1.22
15. I will focus on the important issues during discussion activities	5.32	0.94
16. I will participate in interactive discussions when I read annotations by other readers	5.21	0.88
Overall satisfaction with group interactions	5.28	
Learning environment		
17. I was satisfied with the CRAS-RAIDS as a learning tool and its use for performing reading activities	5.25	0.75
18. I did not feel pressured during reading activities in CRAS-RAIDS	5.04	1.07
19. The time needed to perform reading activities in CRAS-RAIDS is reasonable	5.04	0.96
20. The CRAS-RAIDS provides opportunities to express my thoughts about the article	5.32	0.72
21. The CRAS-RAIDS can facilitate knowledge sharing by readers	5.29	0.85
22. Browsing reading annotations in CRAS-RAIDS was useful	5.38	0.78
23. Overall, CRAS-RAIDS is an effective assisted learning tool for reading activities	5.29	0.81
Overall satisfaction with learning environment	5.23	
Learning system		
24. The CRAS-RAIDS has a user-friendly interface	5.54	0.79
25. The login interface in the CRAS-RAIDS is very easy to use	5.25	1.14
26. The interactive discussion interface in the CRAS-RAIDS for responding reading annotations is user friendly	5.29	1.15
27. The functionalities in the CRAS-RAIDS for responding discussion issues are very easy to use	**5.71**	**0.53**
28. Browsing annotations by my peers in the CRAS-RAIDS are very easy	**5.61**	**0.79**
29. Using the designed functionalities in the CRAS-RAIDS for reading annotated digital texts is very convenient	5.43	0.69
30. Reading the annotated digital texts on the CRAS-RAIDS is very convenient	5.14	1.15
31. I will continue using the CRAS-RAIDS to perform reading activity in the future	**5.50**	**0.79**
Overall satisfaction with learning system	5.43	
Overall satisfaction according to questionnaire results	5.25	0.57

(2007) identified three scaffolds, which in their work are called cognitive, technical and affective, to enhance learning performance. Among the three types of scaffolds, the cognitive scaffold denotes mechanisms or activities related to the development of concept. Clearly, the proposed reading annotation and interactive discussion scaffolds, which simultaneously and effectively promote both direct and inferential reading comprehension, can be regarded as a kind of the cognitive scaffolds. The reading annotation scaffold also helps students select appropriate annotation strategies, and the interactive discussion scaffold facilitates high-level thinking and interactive discussion while discouraging the discussion of irrelevant subjects. These experimental results are consistent with those acquired by Chen et al. (2011), who reported that a scaffolded questioning strategy significantly improves reading comprehension when combined with the use of QR codes associated with mobile technology to deliver scaffolded questions to support students in paper-based reading.

Moreover, Chen, Wang, Chen and Wu's study (2014) applied a C4.5 decision tree, which is a widely used data mining technique, to develop a personalized reading anxiety prediction model (PRAPM) to reduce effectively the reading anxiety of learners while reading English articles based on individual learners' reading annotation behavior in a collaborative digital reading annotation system. Their study indicated that the high-level reading annotations and discussed annotations are more important factors affecting reading comprehension performance. Thus, the effects of promoting high-level and discussed annotations on reading comprehension performance should be future studied. This work found that reading annotation scaffolds not only help learners focus on annotated concepts or subjects, but also to construct their own knowledge and concepts. Interactive discussion scaffolds of annotated digital texts also help learners discuss and review the ideas and thoughts of others. Meanwhile, this work demonstrated that reading annotation and interactive discussion scaffolds can simultaneously promote the levels of reading annotations and discussed annotations in a reading annotation-based collaborative reading environment, thus improving learners' reading comprehension performance.

Additionally, this work found that interactive discussion by the control group focused mainly on quizzing and clarification activities, which are associated with shallow discussion, and the percentage of other type belonging to chats approached 30% (Table 2). Face-to-face discussions can easily digress from topic. Conversely, in addition to the quizzing and clarification types, interactive discussion by the experimental group focused on discrimination and reasoning types, which are strongly related to in-depth discussion, and roughly 10% of interactive discussions was identified as the other type belonging to chats. Clearly, the designed asynchronous interactive discussion scaffold indeed improves discussion quality by providing a clear direction for discussion, by allowing time for reflection before discussion, and by facilitating the sharing of information with other readers, thereby helping to clarify arguments. These investigation results are consistent with those in several studies, confirming that computer-mediated discussion has fewer time and space restrictions and allows more opportunities for learners to prepare, reflect, think, and search for information before participating in a discussion than face-to-face discussion (Chen & Chiu, 2008; Guiller et al., 2008). Moreover, this work demonstrated that interactive discussion scaffolds can direct the directions of discussed annotations in a reading annotation-based collaborative reading environment, thus largely reducing the percentage of the other type belonging to chats.

More importantly, the use of reading strategies was significantly higher in the experimental group than in the control group. Successful readers generally use various strategies to understand texts. Dogan (2002) noted that good readers use many strategies before, during, and after reading. Brown et al. (1981) suggested that the use of an improper reading strategy adversely affects reading effectiveness. Clearly, the proposed reading annotation scaffold benefits students by facilitating their use of appropriate reading strategies and their use of the metacognitive skills needed for effective reading, thereby promoting use of reading strategy. Furthermore, analytical results showed significant improvements in the behavioral and affective sub-dimensions and the total dimension for reading attitude in the experimental group, but not in the control group. However, both groups did not differ significantly in reading attitude after the collaborative reading activities. Our interviews with several elementary school teachers who had over ten years of reading instruction experience in Taiwan revealed that promoting reading attitudes within a short period is very challenging. The various collaborative reading approaches used in this study were limited to a period of two weeks. In future studies, comparing reading attitude among different learning periods may prove fruitful.

Finally, some study limitations merit consideration. First, the effects of the proposed CRAS-RAIDS on reading attitude, reading comprehension, and reading strategy were only assessed in a specific age group of elementary school students. Thus, the research results cannot be transferred readily to other age groups and subjects with different academic levels. Second, this study uses two PIRLS articles to assess the effects of the proposed CRAS-RAIDS on reading attitude, reading comprehension, and reading strategy use. Thus, further research should focus on whether research results can be transferred to other articles.

6. Conclusions and future work

A novel CRAS-RAIDS system was used to facilitate high-level reading comprehension and strategic reading in a collaborative digital reading environment. The effectiveness of the proposed CRAS-RAIDS for promoting reading performance was evaluated by comparing the experimental group students (CRAS-RAIDS) and control group students (conventional paper-based reading annotation and face-to-face discussion) in terms of reading attitude, reading comprehension, and use of reading strategies. Analytical results show that use of the proposed CRAS-RAIDS in collaborative reading can substantially improve interactive discussion and high-level thinking and can reduce the time spent on irrelevant discussions. Most importantly, compared to collaborative reading with traditional paper-based reading annotations and face-to-face discussion, collaborative reading with the proposed CRAS-RAIDS promotes direct and explicit comprehension, inferential comprehension, and reading strategy use. However, the experimental and control groups did not significantly differ in reading attitude. Most students in the experimental group were very satisfied, and their opinions of using the proposed CRAS-RAIDS were consistent. Students in the experimental groups also expressed their intention to continue using the CRAS-RAIDS for online collaborative reading in the future.

Additional studies are warranted. First, further studies can compare junior high school students, senior high school students and college students to determine whether the effectiveness of the CRAS-RAIDS system differs by academic levels. Second, since different article styles (*e.g.*, lyrical, practical, and expositive) reportedly affect student reading performance and self-efficacy (McCabe, Kraemer, Miller, Parmar, & Ruscica, 2006), future studies should investigate how different article styles affect reading comprehension, reading strategy use, and reading attitude. In addition to high-level interactive discussion of annotated digital texts, reading high-level annotations, including summary, discrimination, and reasoning types, also have important positive effects on reading comprehension. Therefore, further study should explore correlations between reading annotation levels and reading comprehension. Another potential research direction is the effects of personal characteristics such as prior knowledge, learning style, metacognitive ability, and reading skill on reading comprehension, reading strategy use, and reading attitude when using the CRAS-RAIDS for online collaborative reading. Additionally, since Yelland and Masters (2007) claimed that a scaffold should be withdrawn gradually as competency increases, further studies are needed to investigate whether the proposed scaffolds can still help students become competent e-readers. Finally, since changing reading attitudes is very challenging, future studies can extend the learning period for online collaborative reading to observe variations in reading attitude under realistic conditions.

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